Diskretna TFS

$$F_k = \frac{1}{N}\sum_{n=0}^{N-1} f[n]\cdot e^{-jk\frac{2\pi}{N}\cdot n} \quad = \text{DTFS}$$

$$f[n] = \sum_{n=0}^{N-1} \underbrace{F_k}_{} \cdot e^{jk\frac{2\pi}{N}n}$$

$$F_k = |F_k|\cdot e^{j\measuredangle F_k}$$

! • napisati za prvih npr 5k pa onda tek napisati da je periodičan

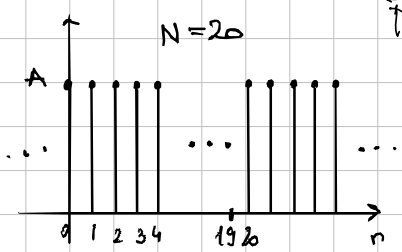
$$F_{k+N} = \frac{1}{N}\sum_{n=0}^{N-1} f[n]\, e^{-j(k+N)\frac{2\pi}{N}n}$$

$$e^{-jk\frac{2\pi n}{N}}\cdot \underbrace{e^{-j2\pi n}}_{1} \implies F_{k+N} = F_k$$

$$P = \frac{1}{N}\sum_{n=0}^{N-1} |f[n]|^2 = \sum_{k=0}^{N-1} |F_k|^2$$

* apsolutne zagrade su da ki signal bio kompleksan

Zadatak:



budući da kreće od 0 ide L-1

$$F_k = \frac{1}{N}\sum_{n=0}^{N-1} f[n]\cdot e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}\sum A e^{-j\frac{2\pi}{N}kn}$$

gledaj se: $\sum_{n=0}^{\infty} g^n = \frac{1}{1-g}$ beskonačan, $\sum_{n=0}^{N-1} g^n = \frac{1-g^N}{1-g}$ konačan

geom red

$$F_k = \frac{A}{N}\sum_{n=0}^{L-1}\left[e^{-j\frac{2\pi}{N}k}\right]^n = \frac{A}{N}\cdot\frac{1-e^{-j\frac{2\pi}{N}k\cdot L}}{1-e^{-j\frac{2\pi}{N}\cdot k}}$$

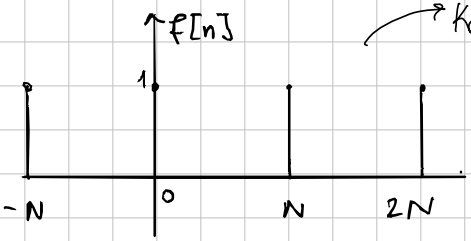
$$= \frac{A}{N}\cdot\frac{e^{-j\frac{2\pi}{N}\cdot k\cdot\frac{L}{2}}\left(e^{j\frac{2\pi}{N}\cdot k\cdot\frac{L}{2}} - e^{-j\frac{2\pi}{N}\cdot k\cdot\frac{L}{2}}\right)}{e^{-j\frac{2\pi}{N}\cdot k\cdot\frac{1}{2}}\left(e^{jk\frac{2\pi}{N}\cdot k\frac{1}{2}} - e^{-j\frac{2\pi}{N}\cdot k\frac{1}{2}}\right)}$$

$$= \frac{A}{N}\cdot e^{-j\frac{2\pi}{N}k\left(\frac{L}{2}-\frac{1}{2}\right)}\,\frac{\cancel{2j}\sin\frac{2\pi}{N}k\cdot\frac{L}{2}}{\cancel{2j}\sin\frac{2\pi}{N}\cdot k\cdot\frac{1}{2}} = \frac{A}{N}\cdot e^{-j\frac{2\pi}{N}\cdot k\left(\frac{L}{2}-\frac{1}{2}\right)}\cdot\frac{\sin\frac{2\pi}{N}\cdot k\cdot\frac{L}{2}}{\sin\frac{2\pi}{N}\cdot k\cdot\frac{1}{2}}$$

Za $k=0$ → u prvi red, ne u zadnji!

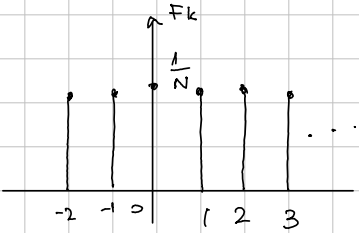
$$F_0 = \frac{A}{N}\sum_{n=0}^{L-1}\left[e^{-j\frac{2\pi}{N}\underset{0}{k}}\right]^n = \frac{A}{N}\sum_{n=0}^{L-1}\underset{1}{1}^n = \frac{A\cdot L}{N}$$

Zadatak 2.)



Kronickerova delta fja: (samo u N)

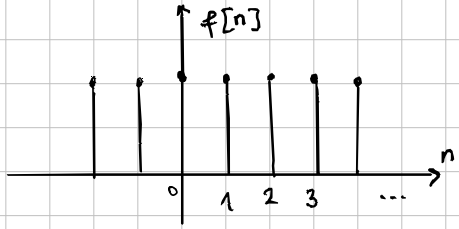
$$F_k = \frac{1}{N}\sum_{n=0}^{N-1} \underbrace{\delta[n]}_{1} e^{-j\frac{2\pi}{N}kn} = \frac{1}{N}$$



$$P = \frac{1}{N}\sum_{n=0}^{N-1} |\delta[n]|^2 = \frac{1}{N}$$

$$P = \sum_{k=0}^{N-1} |F_k|^2 = \sum_{k=0}^{N-1} \left(\frac{1}{N}\right)^2 = \frac{1}{N^2}\sum_{k=0}^{N-1} 1 = \frac{1}{N}$$

Zadatak 3.)



$$F_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi jkn}{N}}$$

$x \rightarrow$ periodičan signal 1

$$F_k = \frac{1}{N}\sum_{n=0}^{N-1}\left(e^{-j\frac{2\pi}{N}k}\right)^n = \frac{1}{N}\,\frac{1-e^{-j\frac{2\pi}{N}k\cdot N}}{1-e^{-j\frac{2\pi}{N}k}} = 0 \;?$$

očito tu nije kraj jer ne može biti 0

$k=0$ će to „poništiti"

$$\rightarrow F_0 = \frac{1}{N}\sum_{n=0}^{N-1} 1\cdot \underbrace{e^0}_{1} = \frac{1}{N}\cdot N = 1$$

$\rightarrow$ ponovi se samo u 1 $\rightarrow$ kronicker δ

$\Rightarrow$

$\rightarrow$ periodičan je za N

$$P = \frac{1}{N}\sum_{n=0}^{N-1} |f[n]|^2 = \frac{1}{N}\sum_{n=0}^{N-1} 1^2 = 1$$

$$P = \sum_{k=0}^{N-1} |F_k|^2 = \sum_{k=0}^{N-1} |\delta(k)|^2 = 1$$